Project Name

Hotel Reservation System

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WORKBOOK

Project Name: Hotel Reservation System

1. Problem Statement

Many hotels still handle room bookings through phone calls, walk-ins and paper registers. This leads to double bookings, outdated room availability, long waiting times for guests, payment errors and no reliable record of reservations, occupancy or revenue.

Our Hotel Reservation System provides a digital platform where guests search for available rooms, make and cancel reservations and pay online, while hotel staff / admin manage rooms, reservations, customer details and reports.

2. Why Does the Problem Exist?

- **Manual Booking** – Reservations are recorded in registers or spreadsheets, which causes errors.
- **Outdated Availability** – Room status is not updated in real time, so rooms get double booked.
- **Poor Communication** – Guests cannot check rooms, rates or confirmations at any time.
- **Payment Issues** – Manual payment handling is slow and difficult to track.
- **Lack Of Records** – There is no central data on guests, bookings, occupancy or revenue.
- **Cancellation Handling** – Cancellations and modifications are not reflected quickly in availability.

3. Internal Stakeholders

- **Hotel Staff / Admin** – manages rooms, reservations, customer details and reports.
- **Hotel Management** – oversees operations, occupancy and revenue.
- **System Administrator / Developer** – maintains the website and database.

4. External Stakeholders

- **Guests / Customers** – search rooms, book, pay and cancel reservations.
- **Payment Gateway / Bank / UPI** – processes card and online payments.
- **Travel Agents** – may book rooms on behalf of guests.
- **Government / Local Authorities** – tourism, tax and guest-registration regulations.

5. Expected System Users

- **Customer / Guest** – registers, logs in, searches rooms, makes or cancels reservations and views reservation details.
- **Hotel Staff / Admin** – logs in, manages rooms (add / update / delete), manages reservations, views customer details and generates reports.

6. Objectives of the Project

- **Online booking** – develop an online hotel reservation system.
- **Room search** – let guests search rooms by date and see real-time availability.
- **Reservations** – allow guests to create, view and cancel reservations.
- **Payments** – support secure payment with a choice of payment method and invoice generation.
- **Admin tools** – let staff manage rooms, reservations and customer information.
- **Records** – maintain accurate reservation, payment and occupancy records.

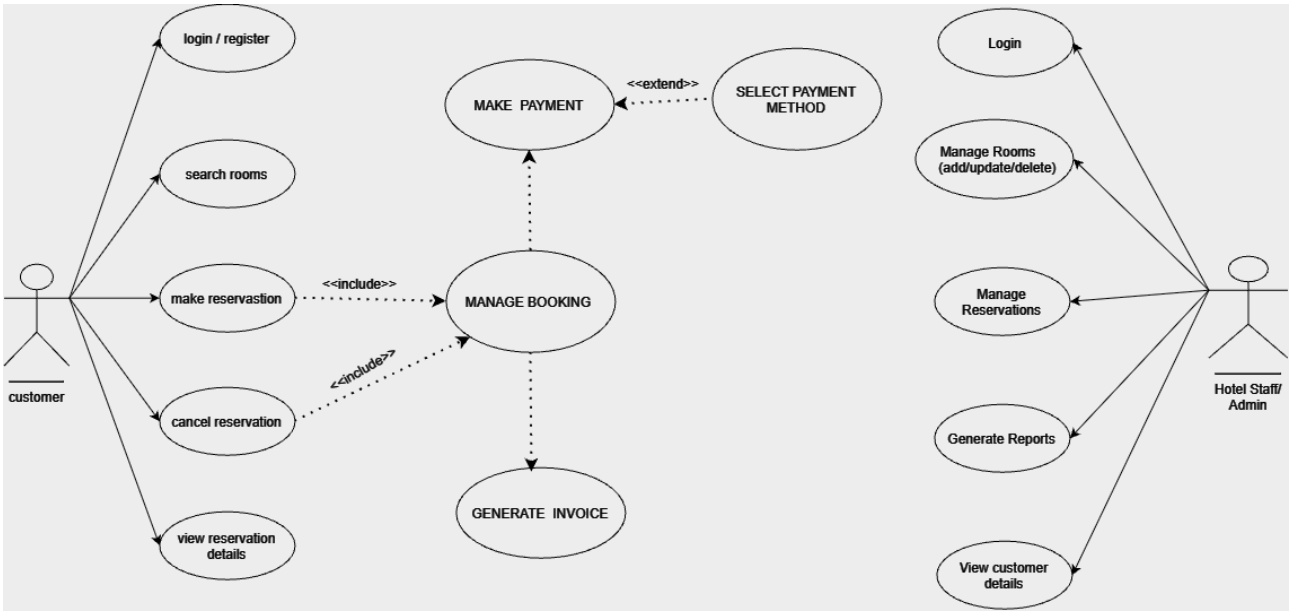
7. Expected Outputs

- **Room listing** – available rooms with type, rate and availability.
- **Reservation** – online reservation with booking confirmation.
- **Payment** – payment confirmation and invoice / receipt.
- **Reservation status** – e.g. Pending, Confirmed, Cancelled.
- **Admin dashboard** – room, reservation and customer management.
- **Reports** – occupancy, bookings and revenue reports.

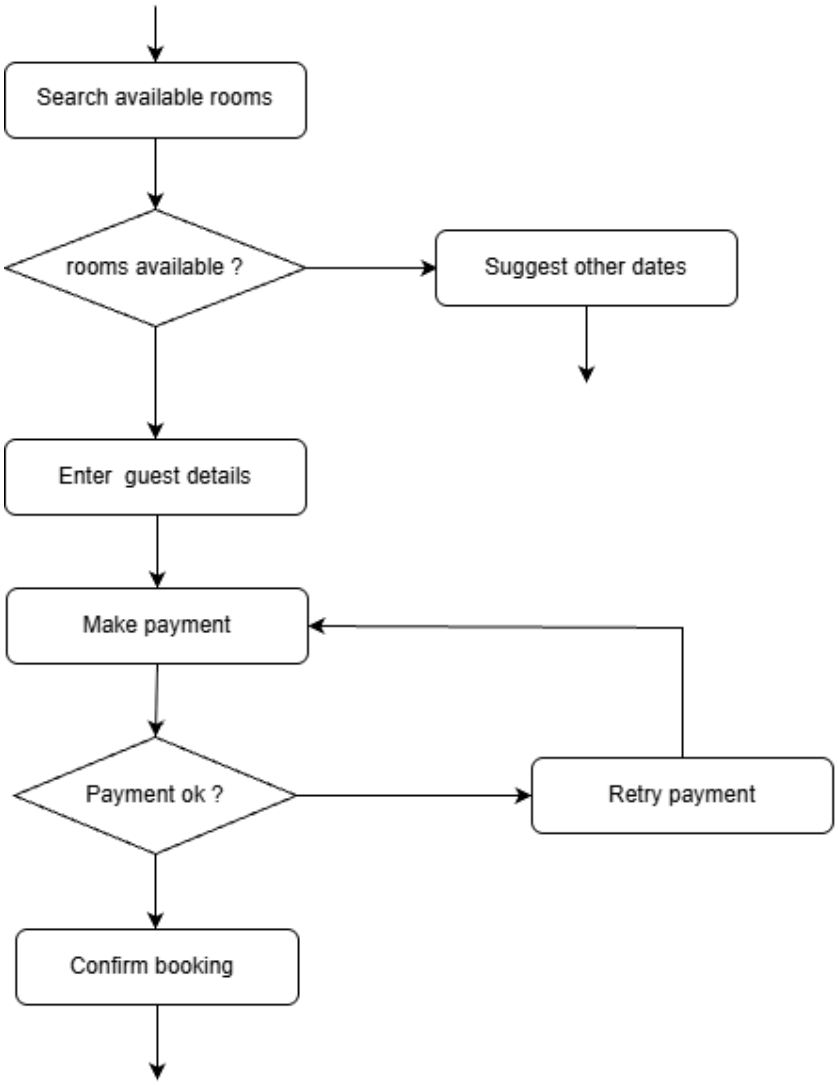
8. Hands-on Activity

- **Survey** – study how hotels handle bookings, rooms and payments.
- **Data gathering** – collect details on guests, rooms, reservations and payments.
- **Wireframe** – design the website interface for search, booking and admin.
- **Workflow** – create the reservation workflow: search, guest details, payment, confirmation.
- **Stakeholders** – identify stakeholders and expected system users.
- **Database** – design tables for guests, staff, rooms, reservations and payments.

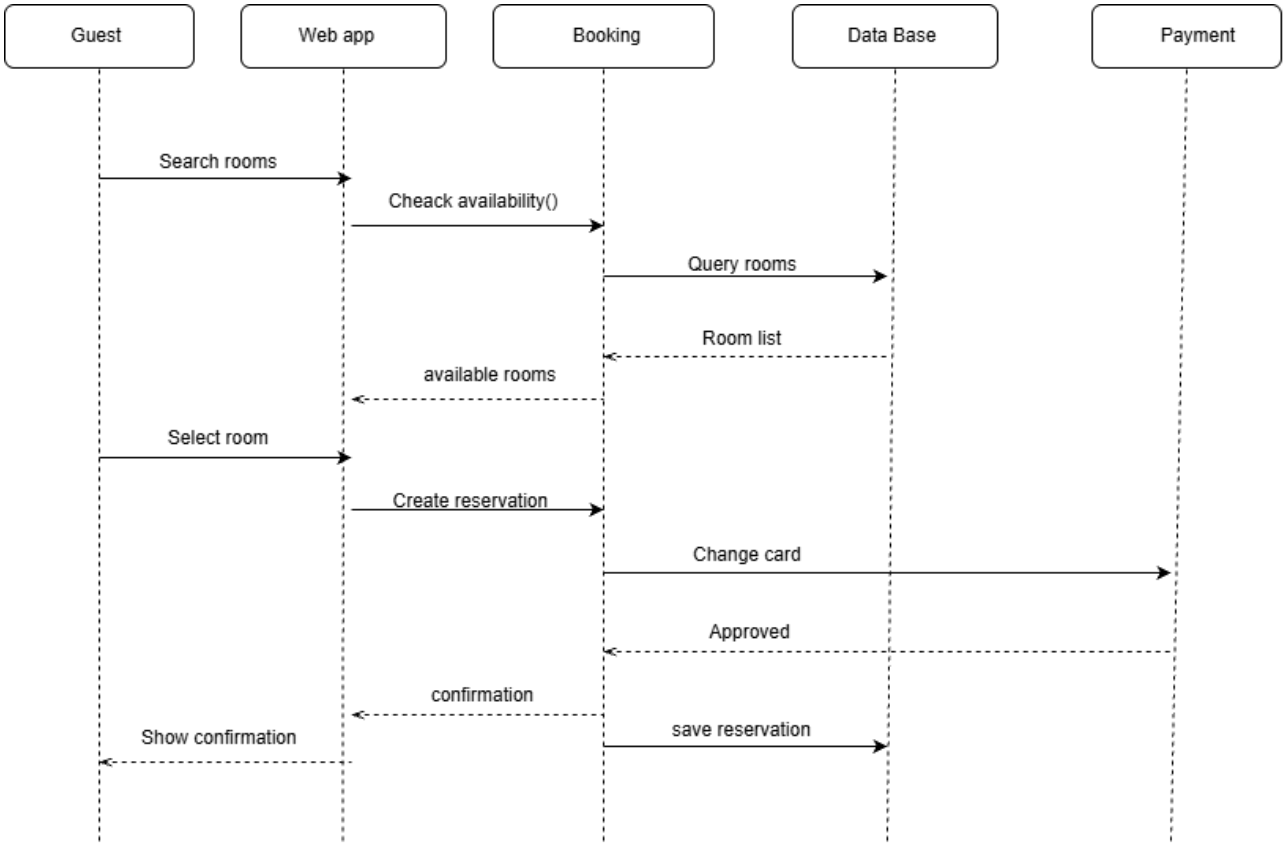
Use Case Diagram - Hotel Reservation System



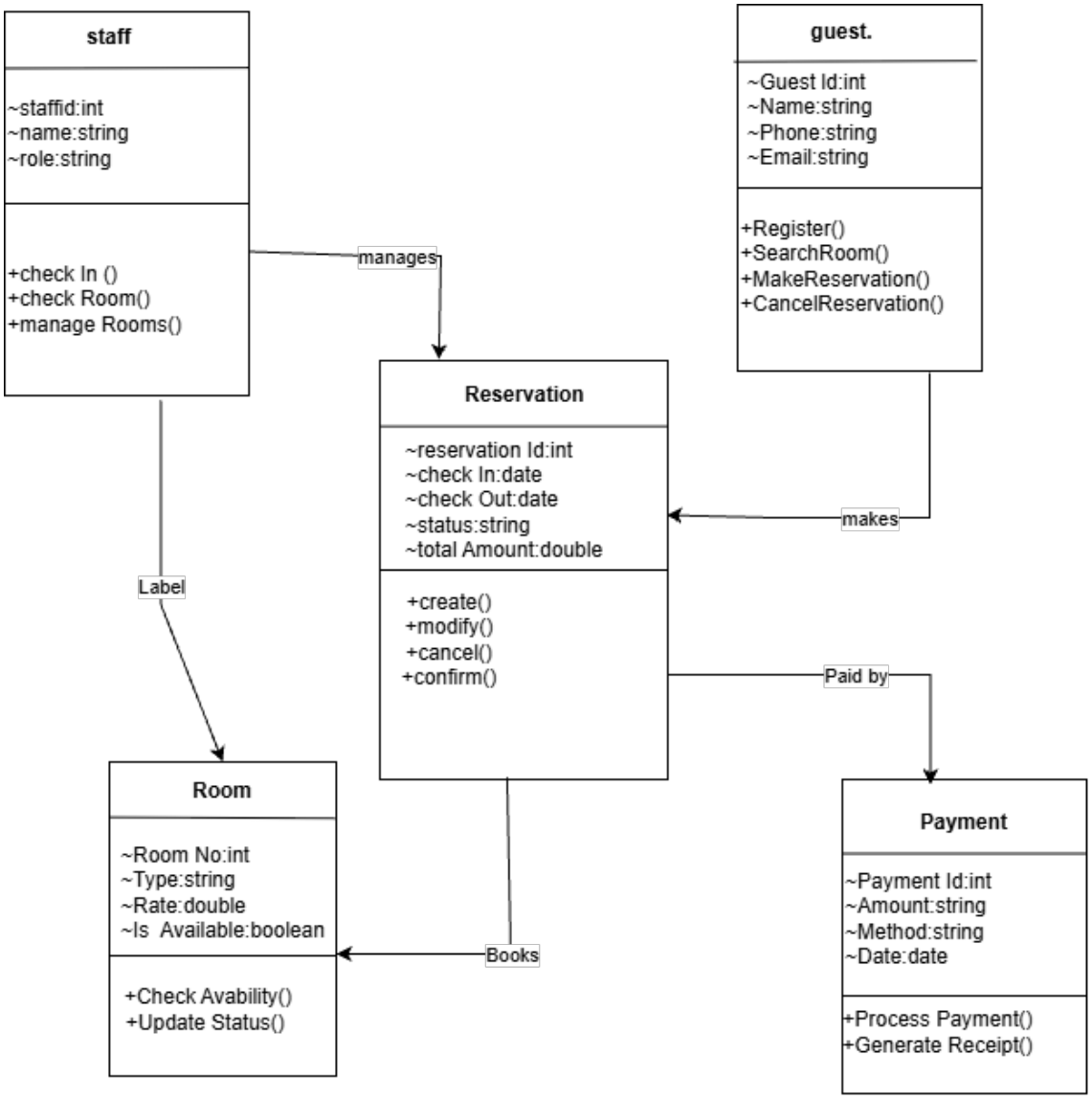
Activity Diagram - Hotel Reservation System



Sequence Diagram - Hotel Reservation System



Class Diagram - Hotel Reservation System



Collaboration Diagram - Hotel Reservation System

